



Boron-containing single-ion conducting polymer electrolyte for dendrite-free lithium metal batteries

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ABSTRACT

Lithium metal is regarded as a “Holy Grail” anode material because of its high theoretical specific capacity and uniquely low reduction potential. However, its low Coulombic efficiency and hazards caused by uncontrolled lithium dendrite growth have hindered its practical application in anodes for many years. In this work, we design a novel boron-containing single-ion conducting polymer electrolyte that eliminates the large electric field that Chazalviel has identified as the cause of lithium dendrite growth. The ability to inhibit lithium dendrite growth is demonstrated via galvanostatic cycling tests of Li/Electrolyte/Li cells. In addition, the boron-containing single-ion conducting polymer electrolyte exhibits excellent ionic conductivity (2.23 mS cm^{-1}) and an extremely high lithium-ion transference number (0.71). Moreover, the $\text{LiFePO}_4/\text{Electrolyte}/\text{Li}$ cell exhibits an outstanding specific capacity of more than 130 mAh g^{-1} and excellent capacity retention (90% of the highest capacity was retained after 100 cycles). This excellent performance indicates a promising strategy for development of safe, long-life lithium metal batteries.

1. Introduction

The explosive growth of electric vehicles, portable electronics, and grid storage have made development of high-energy-density batteries important [1,2]. Lithium metal is thought to be a substantially better anode material than commercially available materials such as graphite due to its ultrahigh theoretical specific capacity (3860 mAh g^{-1}) and uniquely low reduction potential (-3.04 V vs the standard hydrogen electrode) [3,4]. Researchers have sought to apply Li metal batteries (LMBs) to practical applications since the 1970s. However, low Coulombic efficiencies and poor safety performance caused by uncontrollable Li dendrite growth have severely hindered practical applications [5,6].

Li dendrite formation can be divided into two steps: nucleation and growth. All Li dendrite elimination strategies are aimed at suppressing nucleation and growth. As we know, the component distribution and property of solid electrolyte interphase (SEI) have a great influence on the lithium deposition behavior and the growth of lithium dendrites. Hence, a lot of methods modifying and enhancing SEI were proposed to suppress the growth of lithium dendrites [7,8]. Besides, in 1990, space-charge theory was proposed to describe Li dendrite nucleation [9]. It led to another strategy to achieve dendrites-free anode. Chazalviel divided a conventional non-aqueous electrolyte into quasi-neutral and

space-charged areas within a small area near the anode surface. Ion transfer is driven by diffusion in the former region and governed by electric fields in the latter. When Li ions deposit quickly on an anode surface, anion depletion occurs near the anode, leading to a space charge and the ramified growth shown in Fig. 1a [10,11]. Moreover, Li ions with larger solvation clusters are less mobile than counterions. Therefore, concentration polarization occurs during cell cycles, leading to low Coulombic efficiency and performance deterioration [12–14]. Hence, identification of a novel electrolyte that can regulate ion distribution to suppress Li dendrite growth is crucial to LMB development.

In recent years, various methods have been proposed for ion distribution regulation. These include use of electrolyte additives [15], ceramic fillers [16,17], and single-ion conductors [18–21]. Single-ion conducting polymer electrolytes (SIPEs) can achieve particularly high lithium-ion transference numbers. Anion immobilization is the most popular SIPE design strategy. However, most of the resulting materials have suffered from low ionic conductivities due to poor delocalization of polymeric lithium salts. The strategy of introducing anion receptors into electrolytes was recently proposed to improve lithium-ion transference numbers [22,23]. This can improve not only the lithium-ion transference number but also the ionic conductivity because of Lewis acid-base interactions between anions and their receptors.

In this work, we developed a novel SIPE with anion receptors,

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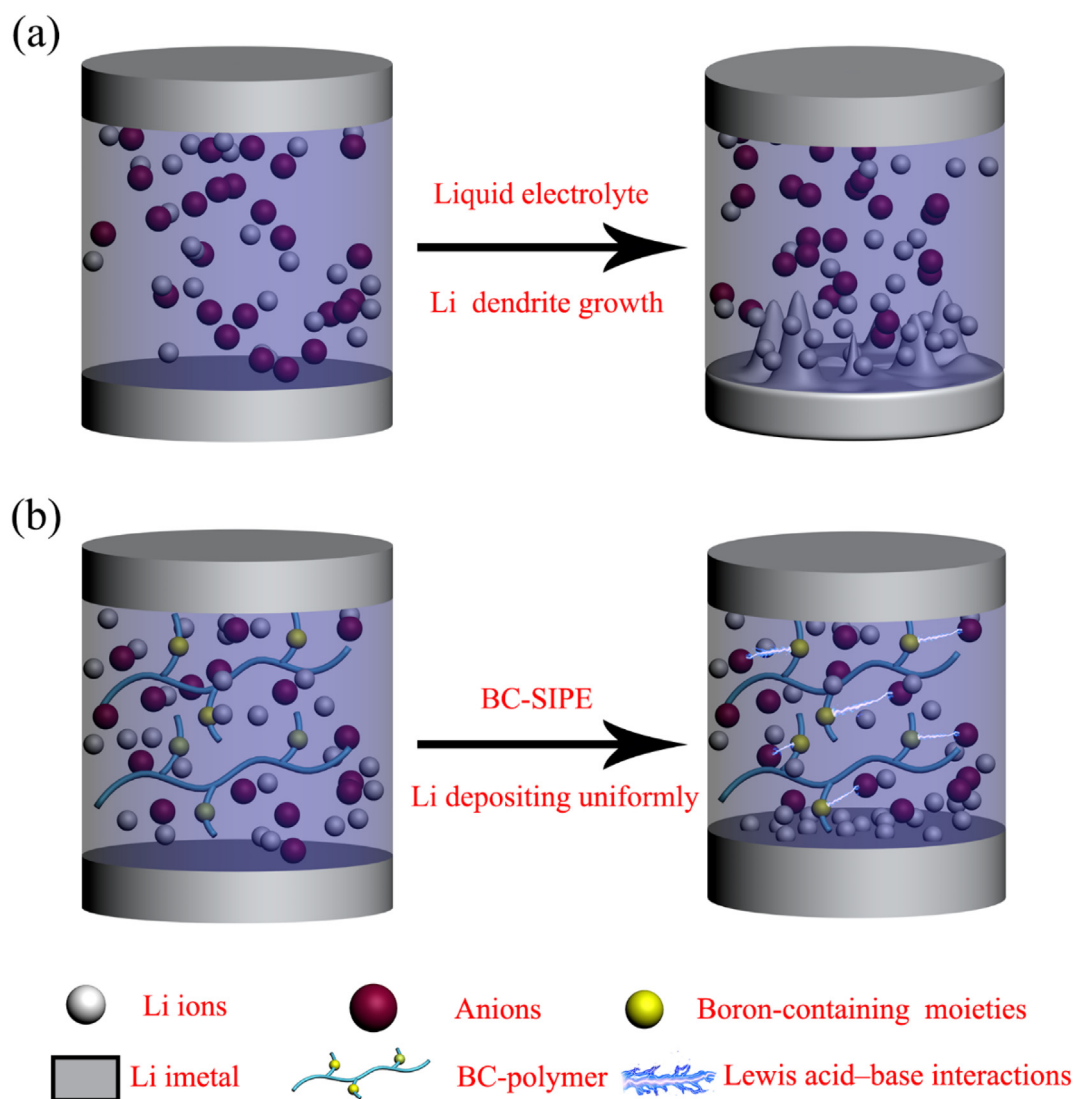


Fig. 1. a) Schematic illustration of Li dendrite growth in a liquid electrolyte. b) Schematic illustration of uniform Li deposition in BC-SIPE.

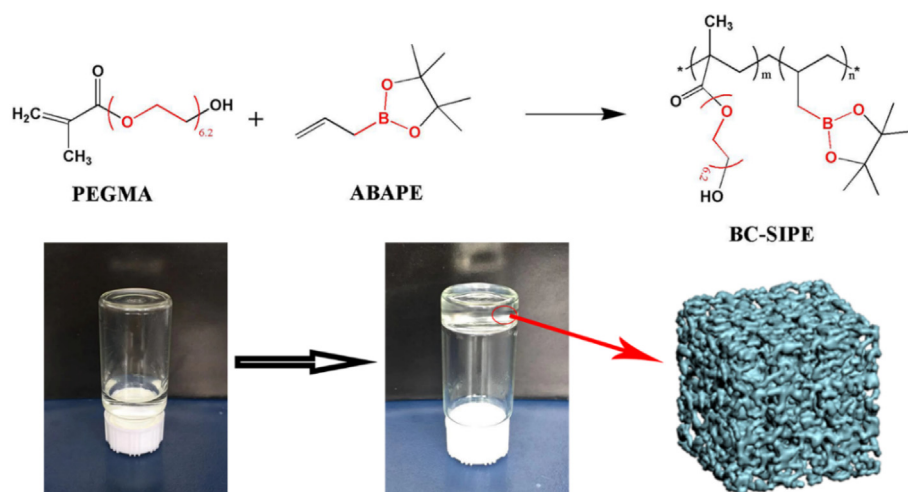


Fig. 2. Schematic illustration of the BC-SIPE polymerization process.

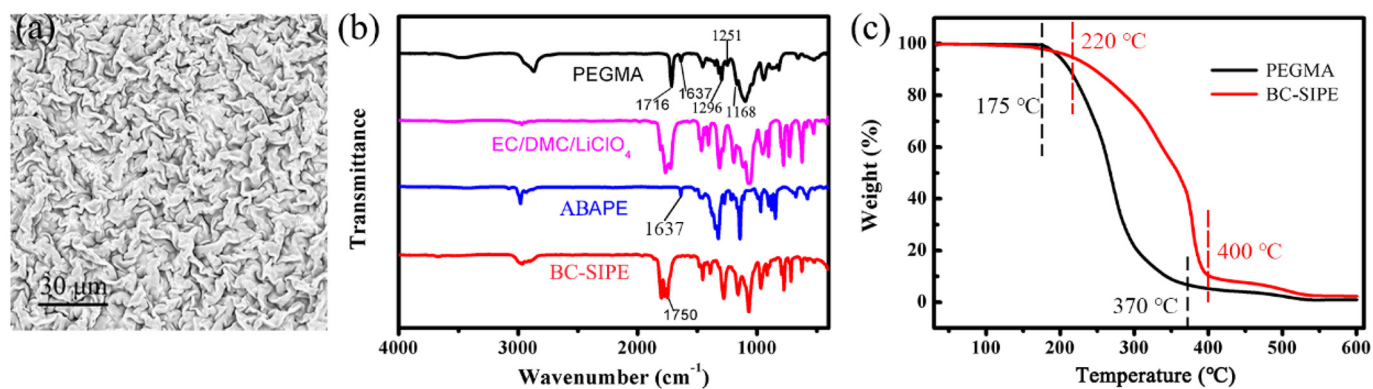


Fig. 3. a) SEM image of the BC-polymer membrane surface. b) FT-IR spectra of PEGMA, EC/DMC/LiClO₄, ABAPE, and BC-SIPE. c) TGA curves of PEGMA and BC-SIPE.

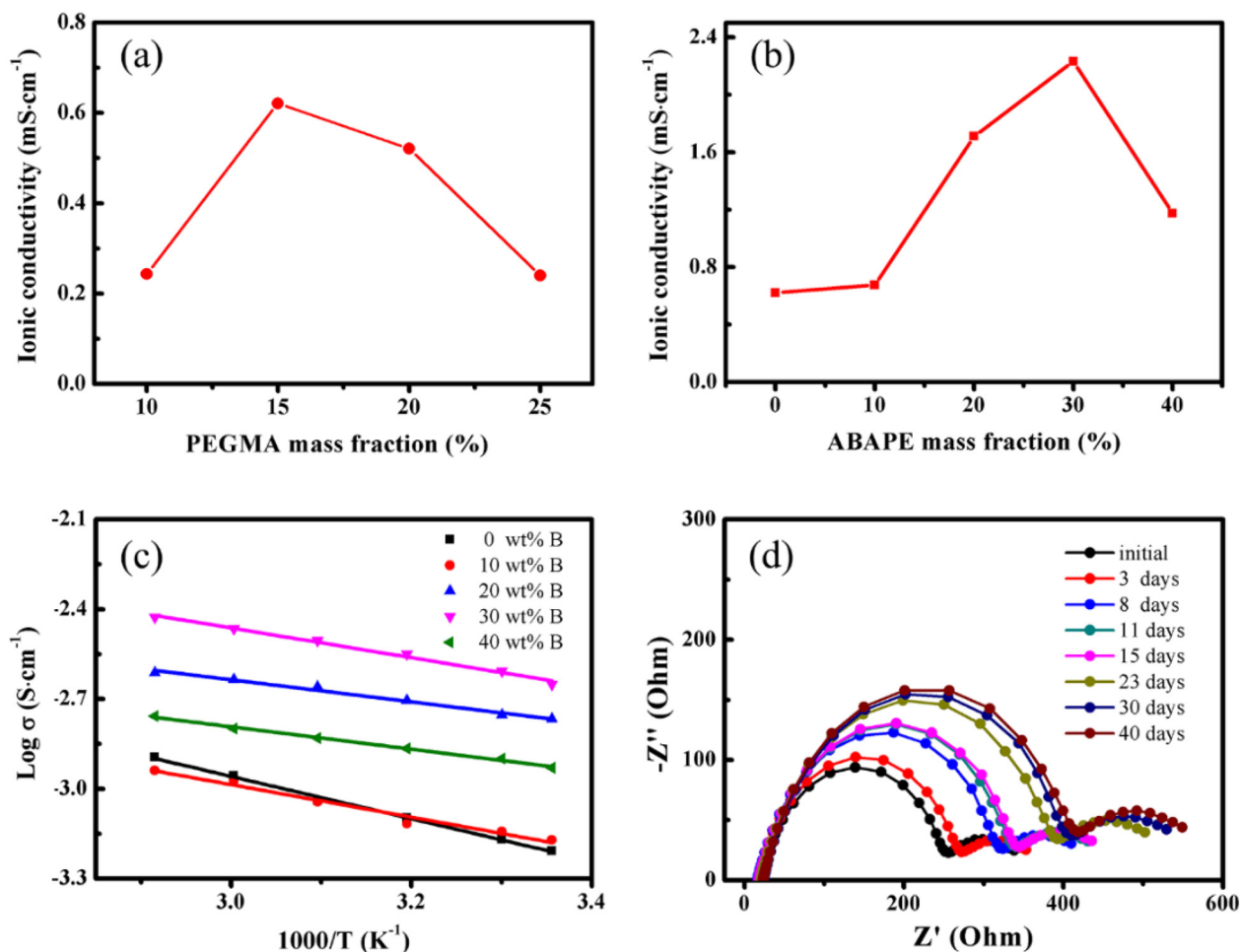


Fig. 4. a) Influence of the PEGMA mass fraction on ionic conductivity; b) Influence of the ABAPE mass fraction on ionic conductivity (PEGMA mass fraction 15%); c) Ionic conductivities of electrolytes made with various ABAPE mass fractions at various temperatures; d) The interfacial resistance of BC-SIPE.

boron-containing moieties, and ethylene oxide (EO) chains. The resulting boron-containing single-ion conducting polymer electrolyte (BC-SIPE) exhibited outstanding electrochemical performance with high ionic conductivity (2.4 mS cm^{-1}), and a large lithium-ion transference number (0.71). This excellent performance ensured that the electrolyte could inhibit lithium dendrite growth based on the theory by Chazalviel. Galvanostatic cycling tests of Li/BC-SIPE/Li cells indicated stable overpotentials even after 500 h at a current density of 0.2 mA cm^{-2} . Moreover, the LiFePO₄/BC-SIPE/Li cell exhibited excellent cycling performance. The discharge capacity was 135.9 mAh

g^{-1} at 0.2 C after 100 cycles, which is 90% of the maximum capacity.

2. Results and discussion

2.1. BC-SIPE preparation and characterization

BC-SIPE was synthesized via free radical polymerization with PEGMA and ABAPE, as shown in Fig. 2. After the thermally initiated polymerization, a piece of flexible film made of PPEGMA and boron-containing polymer (BC-polymer) is formed, as shown in Fig. S1. An

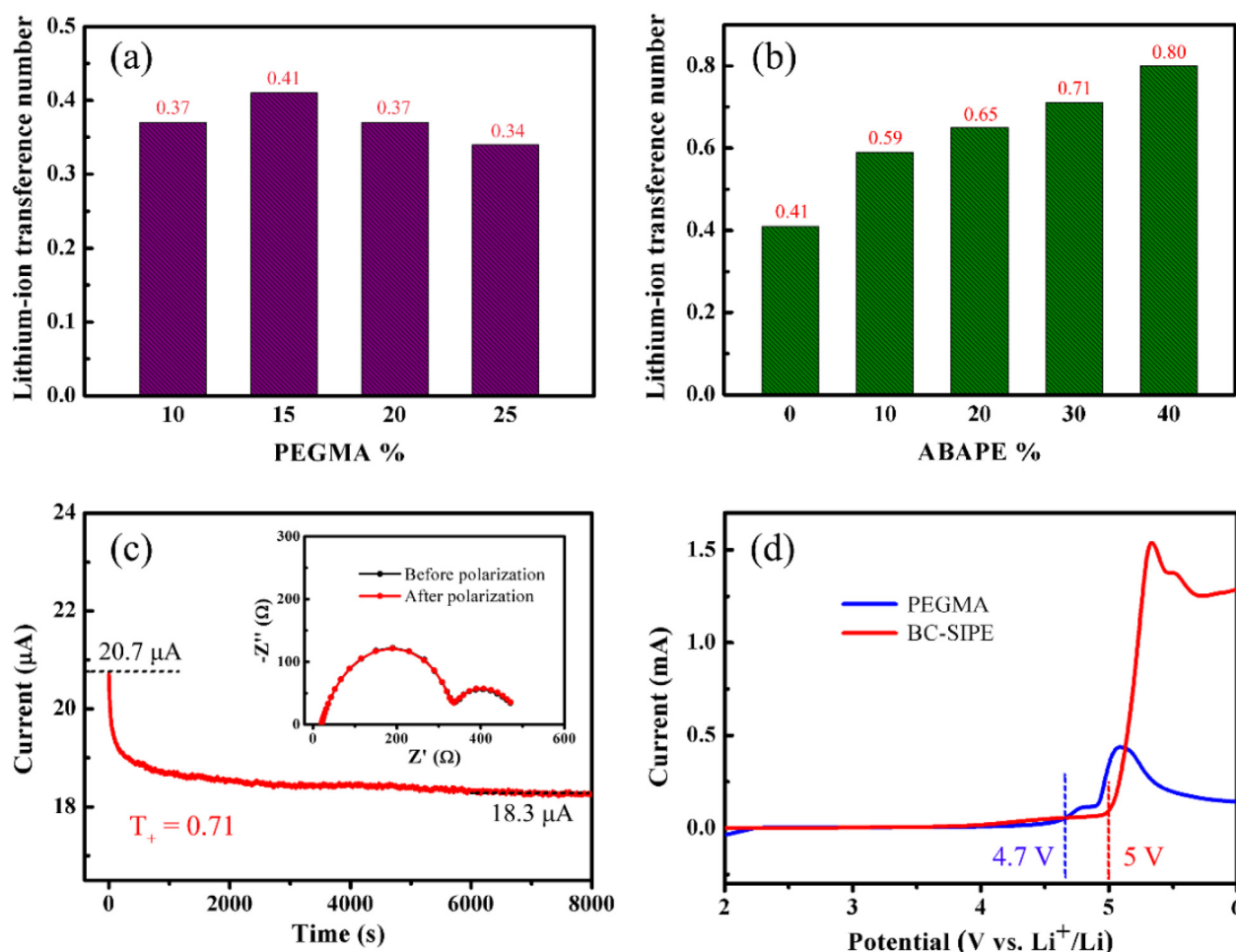


Fig. 5. a) Influence of the PEGMA mass fraction on the lithium-ion transference number; b) Influence of the ABAPE mass fraction on the lithium-ion transference number (PEGMA mass fraction 15%); c) CA and EIS of the Li/BC-SIPE/Li symmetric cell (PEGMA and ABAPE mass fractions 15% and 30%, respectively); d) PEGMA and BC-SIPE linear sweep voltammetry results.

SEM image of the polymer surface is shown in Fig. 2a. The membrane surface is covered with pleats. The FT-IR spectra of PEGMA, EC/DMC/ LiClO_4 , ABAPE, and BC-SIPE are shown in Fig. 2b. The peak at 1637 cm^{-1} corresponds to the C=C stretching vibration [24,25] and is present in PEGMA and ABAPE, but not in BC-SIPE. This indicates that polymerization is complete and almost no monomer remains. In addition, the peak at 1716 cm^{-1} is the characteristic C=O peak of an acrylic ester. After polymerization, this peak moves to 1750 cm^{-1} in BC-SIPE due to disappearance of the conjugative effect. The peaks at 1296 cm^{-1} , 1251 cm^{-1} , and 1168 cm^{-1} correspond to C-O-C stretching and bending vibration peaks in PEGMA [26,27].

High thermal stability is one important reason for application of polymer electrolytes to LMBs. To investigate the thermal stability of BC-SIPE, TGA measurements of BC-SIPE and pure PPEGMA are shown in Fig. 2c. In pure PPEGMA, weight loss begins at 175°C and continues until 370°C (approximately 92% weight loss). The material is degraded completely at 520°C . In contrast, the temperature of initial decomposition increases from 175°C to 220°C after introducing ABAPE. In addition, until 400°C , the first thermal degradation finished until 400°C (approximately 90% weight loss). Thus, the introduction of ABAPE improves thermal stability.

The ionic conductivities of most SIPEs are too low to be practical. Therefore, the ionic conductivity is a critical factor in SIPE evaluation. EO segment motion can promote conduction of lithium ions. Thus, we have explored the effect of PEGMA concentration on ionic conductivity. As shown in Fig. 4a, the ionic conductivity increases from 0.24 mS cm^{-1}

cm^{-1} to 0.62 mS cm^{-1} due to the increase in EO segment concentration that occurs when the PEGMA concentration increases from 10% to 15%. However, further increasing the PEGMA concentration from 15% to 25% decreases the ionic conductivity of polymer electrolyte from 0.62 mS cm^{-1} to 0.24 mS cm^{-1} due to of the reduced plasticizer content. Hence, we chose the membrane with a PEGMA concentration of 15% for further study.

The influence of the ABAPE content in the monomer on the ionic conductivities of BC-SIPEs with 15% PEGMA was investigated. As shown in Fig. 3b, the ionic conductivity increases from 0.62 mS cm^{-1} to 2.23 mS cm^{-1} when the ABAPE mass fraction increases from 0 to 30%. This is because introduction of ABAPE destroys the regularity of the PPEGMA chain, leading to an increase in the proportion of amorphous regions. However, further increasing the ABAPE mass fraction from 30% to 40% decreases the ionic conductivity to 1.18 mS cm^{-1} . Such behavior is thought to occur because the high polymer concentration hinders movement of Li ions. The temperature dependences of the ionic conductivities of PPEGMA and BC-SIPEs are shown in Figs. S2 and 4c, respectively. EIS tests of SS/BC-SIPE/SS cells were carried out from 25°C to 70°C . The logarithm of the ionic conductivity ($\log \sigma$) decreases linearly with the increase in $1000/T$, indicating that ionic conductivity follows Arrhenius behavior in these cases.

The time dependence of the BC-SIPE interfacial resistance is shown in Fig. 4d. The initial interfacial resistance is only $235\ \Omega$, which is substantially lower than those of gel polymer electrolytes (GPE) prepared via ex-situ polymerization. The impedance increases slowly until

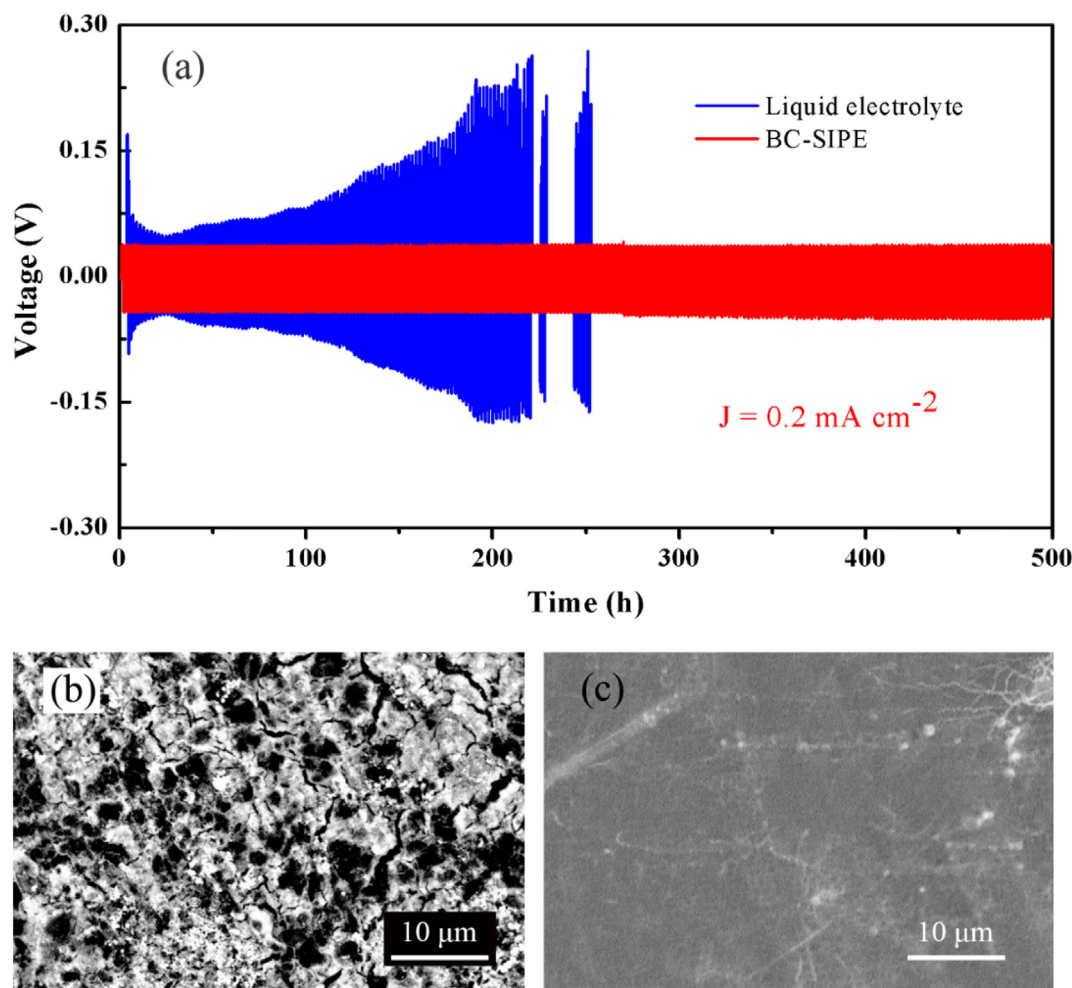


Fig. 6. a) Galvanostatic charge-discharge performance of Li/liquid electrolyte/Li and Li/BC-SIPE/Li cells at a current density of 0.2 mA cm^{-1} . b) and c) SEM images of lithium metal surfaces in the Li/liquid electrolyte/Li cell after cycling for 250 h and the Li/BC-SIPE/Li cell after cycling for 500 h, respectively.

the 23rd day. At this point, the interfacial resistance becomes stable at 390Ω , indicating stable solid electrolyte interface (SEI) film formation.

According to space-charge theory, the Li-ion transference number is a key factor in inhibiting lithium dendrites. The influence of the PEGMA concentration on the Li-ion transference number is shown in Fig. 5a. Moreover, the CAs and EISs of electrolytes with various PEGMA concentrations are shown in Fig. S3. The Li-ion transference numbers of four electrolytes with different PEGMA concentrations are between 0.3 and 0.4, and therefore similar to those of liquid electrolytes. This means that the PEGMA concentration has little effect on the Li-ion transference number. We also explored the impact of the ABAPE mass fraction on the Li-ion transference number. As shown in Fig. 5b, the Li-ion transference number increases rapidly from 0.41 to 0.8 when the ABAPE mass fraction increases from 0 to 40%. Such behavior can be explained via Lewis acid-base interactions between boron moieties in the polymer and anions in the lithium salt. Increasing the quantity of boron moieties in the polymer increases the Li-ion transference number of the electrolyte. However, the precursor solution cannot form a gel regardless of how long it is heated when the ABAPE mass fraction in monomer exceeds 50%. CA and EIS plots of BC-SIPE with 30% ABAPE are shown in Fig. 5c. Plots made using BC-SIPE with 10%, 20%, and 40% ABAPE are shown in Fig. S4. Obviously, the impedances of electrolytes that contain ABAPE are much lower than those of electrolytes that contain only PEGMA. Thus, introducing ABAPE can improve interfacial compatibility between the PEGMA electrolyte and the lithium metal electrode. After considering ionic conductivities and Li-ion transference numbers, the BC-SIPE with 15% PEGMA and 30% ABAPE was chosen for further

study.

Electrolyte electrochemical stability was measured via LSV (Fig. 5d). There is no significant oxidation current in the PEGMA electrolyte until 4.7 V, which indicates an electrochemical window of approximately 4.7 V. In contrast, the electrochemical window of BC-SIPE (5 V) is wider than that of PEGMA electrolyte. This can be reasonably explained as the effect of strong interactions between boron moieties and anions enhancing the electrolyte electrochemical stability. This excellent performance makes the PEGMA electrolyte suitable for high-voltage lithium batteries. The electrochemical performance of LFP/BC-SIPE/Li is shown in Fig. S5. The oxidation and reduction peaks appear at approximately 3.8 V and 3.1 V, respectively. There is only one pair of redox reaction peaks. This indicates that BC-SIPE is electrochemically stable below 4.0 V.

2.2. Lithium dendrites

Symmetric Li/Electrolyte/Li cells were assembled to measure the interfacial stability of the electrolyte/Li combination and voltage polarization performance (Fig. 6). The current density used during cycling was 0.2 mA cm^{-2} . Fig. 6a shows that voltage polarization of the Li/Liquid electrolyte/Li cell starts slightly high at 160 mV and then decreases rapidly. This may contribute to the form of the SEI membrane. Later, the overpotential grows, eventually reaching 280 mV. This can be explained by the destruction of the SEI membrane and the generation of death Li. The strongest evidence of this is that the surface morphology of Li metal includes bumps and hollows after cycling (Fig. 6b). In

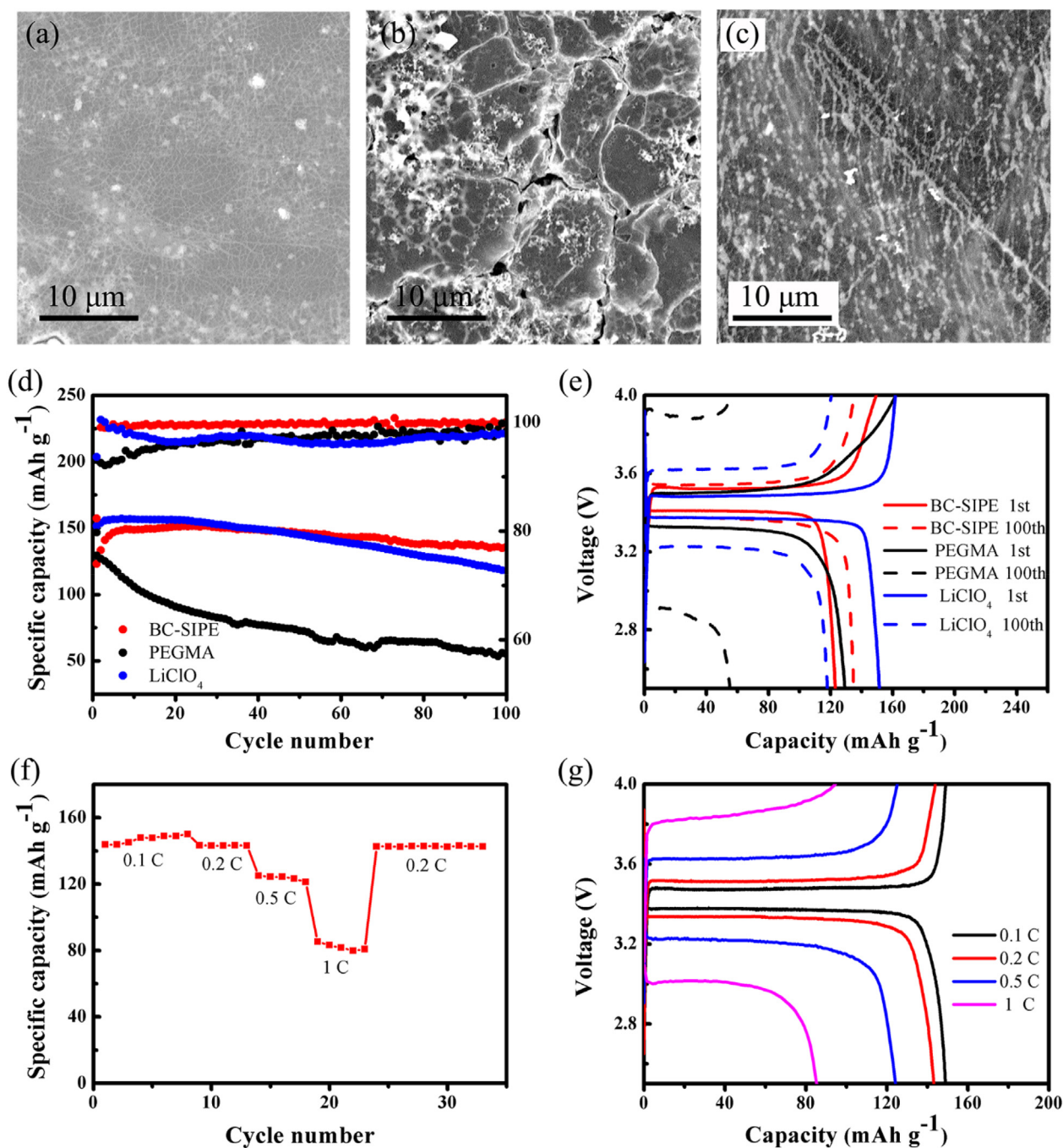


Fig. 7. SEM images of Li anode surfaces from the a) LiFePO₄/BC-SIPE/Li, b) LiFePO₄/PEGMA/Li, and c) vLiFePO₄/LiClO₄/Li cells after 100 cycles; d) Cycling performance of LiFePO₄/BC-SIPE/Li, LiFePO₄/PEGMA/Li, and LiFePO₄/LiClO₄/Li cells at 0.2 C and room temperature; e) Charge-discharge curves at 0.2 C; f) LiFePO₄/BC-SIPE/Li cell rate capability; g) Charge-discharge curves measured at various rates.

contrast, the voltage polarization of the Li/BC-SIPE/Li cell remains stable at 38 mV in throughout cycling. This result is substantially lower than those from Li/Liquid electrolyte/Li cells and other electrolytes. Moreover, there is no obvious dendritic Li on the surface of an anode that has been cycled for 500 h. We can explain this phenomenon via the theory proposed by Chazalviel et al. and described vividly in Fig. 1. When free anions are present, Li ions deposit freely on the surface of the Li metal, resulting in ramified growth. However, in BC-SIPE, the anions in the lithium salt cannot move because of Lewis acid-base interactions between boron-containing moieties and anions. Therefore, there are no large electric fields caused by an anion depletion layer near the Li anode and Li deposition is dendrite-free. As shown in Fig. 6c, the Li

metal surface remains smooth after cycling for 500 h.

2.3. Battery performance

The LFP/BC-SIPE/Li cell was assembled to evaluate BC-SIPE performance in LMBs. In order to better demonstrate the superiority of the BC-SIPE battery, LFP/PEGMA/Li and LFP/LiClO₄/Li cell performance are shown in Fig. 7. After 100 cycles, the discharge capacity of the LFP/BC-SIPE/Li cell is 135.9 mAh g⁻¹ at 0.2C (1C = 170 mAh g⁻¹). This is 90% of its highest capacity. The Coulombic efficiency remains above 99% throughout cycling. Furthermore, the surface of the Li anode remains smooth after cycling (Fig. 7a). In contrast, neither LFP/PEGMA/

Li nor LFP/LiClO₄/Li cell performance is as good as that of the LFP/BC-SIPE/Li cell. Although the initial discharge capacity of the LFP/PEGMA/Li cell is the same as that of the LFP/BC-SIPE/Li cell, the capacity of the former decays much faster than that of the latter. After 100 cycles, the LFP/PEGMA/Li cell discharge capacity is only 55.2 mAh g⁻¹, or 43% of its highest capacity. The lithium metal surface becomes rough after cycling, indicating substantial lithium dendrite formation. The discharge capacity of the LFP/LiClO₄/Li cell also decays quickly, achieving only 75% retention after 100 cycles. This occurs because lithium dendrites grow on the Li metal surface (Fig. 6c). Furthermore, the Coulombic efficiencies of the LFP/PEGMA/Li and LFP/LiClO₄/Li cells are both lower than that of the LFP/BC-SIPE/Li cell. The discharge-charge profiles of these three electrolytes (Figs. S6 and 7e) further demonstrate superior LFP/BC-SIPE/Li cell cycling performance. After 100 cycles, the overpotential values of the LFP/PEGMA/Li and LFP/LiClO₄/Li cells increase from 0.17 V to 1 V and 0.11 V to 0.4 V, respectively. These cells exhibit large polarization increases that are consistent with formation of rough, broken layers on the Li anode (Fig. 7b and c). In comparison, polarization of the LFP/BC-SIPE/Li cell increases only from 0.12 V to 0.16 V (Figs. 7e and S6d). The lithium metal surface remains intact and smooth even after 100 cycles (Fig. 7a).

The rate performance of the LFP/BC-SIPE/Li cell is shown in Fig. 7f. The discharge capacity is 143.7 mAh g⁻¹ at 0.1 C during the initial cycle and reaches 148 mAh g⁻¹ after three cycles. This may be caused by activation during the first few cycles. Moreover, the discharge capacities at 0.2 C, 0.5 C, and 1 C are 143.3, 124.3, and 83.1 mAh g⁻¹, respectively. When the current density returns to 0.2C, the discharge capacity recovers to 142.8 mAh g⁻¹ (99.7% of the initial capacity). This indicates excellent recovery performance. Charge-discharge profiles at various C-rates are shown in Fig. 7d. They are typical of LFP batteries with long, stable charging and discharging platforms. The polarizations are small at 0.1 V and 0.18 V when the current densities are 0.1 and 0.2C, respectively. The excellent BC-SIPE battery performance indicates enormous potential for LMB applications.

3. Conclusions

We successfully prepared a novel BC-SIPE with the ability to limit anion mobility and better thermal stability than PPEGMA. In addition, the optimized BC-SIPE exhibited an excellent ionic conductivity of 2.23 mS cm⁻¹, outstanding electrochemical window of 4.7 V, and high Li-ion transference number of 0.71. Importantly, the ability to suppress Li dendrite growth was demonstrated via stable galvanostatic charge-discharge curves and smooth Li anode surfaces after cycling. Furthermore, the LFP/BC-SIPE/Li cell provided better cycling performance than the LFP/PEGMA/Li or LFP/LiClO₄/Li cells, with 90% retention of its maximum capacity after 100 cycles and high Coulombic efficiency of over 99%. The LFP/BC-SIPE/Li cell also exhibited outstanding rate capabilities (148 mAh g⁻¹, 143.3 mAh g⁻¹, 124.3 mAh g⁻¹, and 83.1 mAh g⁻¹ at 0.1 C, 0.2 C, 0.5 C, and 1 C, respectively). Introducing boron-containing moieties to limit anion movement provides a new approach to inhibiting Li dendrite growth. It also helps to promote practical applications of safe, long-lived LMBs.

Declaration of competing interest

This is a manuscript prepared by Guangping Chen, Chaoqun Niu, Xiaoxue Liao, Yubing Chen, Wenyan Shang, Jie Du* and Yong Chen*, titled "Boron-Containing Single-Ion ConductinLg Polymer Electrolyte for Dendrite-Free Lithium Metal Batteries". It is submitted to be

considered for publication in "Solid State Ionics".

The authors declare that they have no financial and personal relationships with other people or organizations that can inappropriately influence this work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssi.2020.115309>.

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